# Chengpeng Wang

Postdoctoral Fellow in Computer Science, Purdue

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## RESEARCH INTEREST

Software Engineering, Programming Languages, System Security, with a focus on bridging formal methods and machine learning to enhance the reliability and performance of software systems.

## EDUCATION

**Hong Kong University of Science and Technology** *Aug 2019 - Dec 2023*
Ph.D. in Computer Science and Engineering
Committees: *Charles Zhang (advisor)*, *Xiangyu Zhang*, *Shing-Chi Cheung*, *Shuai Wang*
Dissertation: *Enhancing Reliability and Performance of Data-Centric Systems with Static Analysis*

**Tsinghua University** *Aug 2016 - Jul 2019*
M.S. in Software Engineering

**Tsinghua University** *Aug 2012 - Jul 2016*
B.S. in Software Engineering, Minor in Math

## ACADEMIC AND INDUSTRY APPOINTMENTS

Now – Mar 2024 | **Postdoctoral Fellow, West Lafayette, IN, USA**
Neuro-symbolic code auditing, working with Prof. Xiangyu Zhang

July 2023 – May 2023 | **Research Intern at Veridise, Remote**
Static bug detection of zero-knowledge proof circuits

Aug 2022 – Feb 2021 | **Research Intern at Ant Group, Shenzhen, China**
Performance optimization of database-backed application

Aug 2019 – Feb 2019 | **Research Intern at Sourcebrella, Shenzhen, China**
Inconsistency detection of build systems

## HONORS AND AWARDS

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| Postdoc Travel Award, Purdue | *2024* |
| ACM SIGARCH Best Paper Award, ASPLOS'24 | *2024* |
| UGC Research Travel Grant, HKUST | *2022, 2023* |
| Ant Group Outstanding Collaboration Award | *2023* |
| ACM SIGPLAN Distinguished Paper Award, OOPSLA'22 | *2022* |
| ACM SIGPLAN PAC Award | *2022* |
| Outstanding Tutor Award, Student Learning and Development Center, THU | *2019* |
| Future Scholar Award, THU | *2016* |
| Scholarship for Academic Excellence, Tsinghua University | *2013, 2014* |

## GRANTS

PI. "*Neuro-Symbolic Vulnerability Detection for Reliable Software Systems*". Submitted to ***ARC Discovery Early Career Researcher Award Scheme***.

## PUBLICATIONS

**Highlight:** I authored 6 first-author papers and 2 co-first-author papers (labeled with #) published in top-tier conferences and journals, including **ICSE**, **FSE**, **ASE**, **TOSEM**, **OOSPLA**, **ICML**, **NeurIPS**, and **EMNLP**. Additionally, I authored 5 corresponding-author-papers (labeled with $*$) published in **ICSE**, **ASE**, **TOSEM**, and **TSE**. The underlined first authors are Ph.D. students under my mentoring.

## PAPERS IN PROGRESS

[P1] Mingwei Zheng, **Chengpeng Wang**, Xuwei Liu, Jinyao Guo, Shiwei Feng, and Xiangyu Zhang, RFC-Scan: An LLM Agent for Functional Bug Detection in Network Protocols. (Under Review)

[P2] Chao Wang, Li Lin, **Chengpeng Wang**, Jiafeng Huang, Congxia Wu, and Rongxin Wu, ReachCheck: Compositional Library-Aware Call Graph Reachability Analysis in the IDEs. (Under Review)

[P3] Yifei Gao, **Chengpeng Wang**, Pengxiang Huang, Xuwei Liu, Mingwei Zheng, and Xiangyu Zhang, $PR^2$: Peephole Raw Pointer Rewriting with LLMs for Translating C to Safer Rust. (Under Review)

## JOURNAL PUBLICATIONS

[J1] Hao Ling, Heqing Huang, **Chengpeng Wang**, Yuandao Cai, and Charles Zhang, GiantSan: Efficient Operation-Level Memory Sanitization with Segment Folding, In **TOCS 2024**: The ACM Transactions on Computer Systems. (Invited extension of [C10])

[J2] Rongxin Wu, Zhiling Huang, Zige Tian, **Chengpeng Wang**$^*$, and Xiangyu Zhang, PackHunter: Recovering Missing Packages for C/C++ Projects, In **TSE 2024**: Transactions on Software Engineering

[J3] Yi Sun, **Chengpeng Wang**, Gang Fan, Qingkai Shi, and Xiangyu Zhang, Fast and Precise Static Null Exception Analysis with Synergistic Preprocessing, In **TSE 2024**: Transactions on Software Engineering

[J4] Wensheng Tang, Dejun Dong, Shijie Li, **Chengpeng Wang**$^*$, Peisen Yao, Jinguo Zhou, and Charles Zhang, Octopus: Scaling Value-Flow Analysis via Parallel Collection of Realizable Path Conditions, In **TOSEM 2023**: Transactions on Software Engineering and Methodology

[J5] **Chengpeng Wang**, Wenyang Wang, Peisen Yao, Qingkai Shi, Jinguo Zhou, Xiao Xiao, and and Charles Zhang, Anchor: Fast and Precise Value-Flow Analysis for Containers via Memory Orientation, In **TOSEM 2022**: Transactions on Software Engineering and Methodology

## CONFERENCE PUBLICATIONS

[C1] Jinyao Guo$^{\#}$, **Chengpeng Wang**$^{\#}$, Xiangzhe Xu, Zian Su, and Xiangyu Zhang. RepoAudit: An Autonomous LLM-Agent for Repository-Level Code Auditing. In **ICML 2025**: International Conference on Machine Learning.

[C2] Yuan Li, Peisen Yao, Kan Yu, **Chengpeng Wang**, Yaoyang Ye, Song Li, Meng Luo, Yepang Liu, and Kui Ren. Understanding Industry Perspectives of Static Application Security Testing Evaluation, In **FSE 2025**: International Conference on the Foundations of Software Engineering.

[C3] Mingwei Zheng, Danning Xie, Qingkai Shi, **Chengpeng Wang**$^*$, and Xiangyu Zhang, Validating Network Protocol Parsers with Traceable RFC Document Interpretation, In **ISSTA 2025**: International Symposium on Software Testing and Analysis.

[C4] Wei Chen, Bowen Zhang, **Chengpeng Wang**, Wensheng Tang, and Charles Zhang, Seal: Towards Diverse Specification Inference for Linux Interfaces from Security Patches, In **EuroSys 2025**: European Conference on Computer Systems

[C5] **Chengpeng Wang**, Wuqi Zhang, Zian Su, Xiangzhe Xu, Xiaoheng Xie, and Xiangyu Zhang, LLMDFA: Analyzing Dataflow in Code with Large Language Models, In **NeurIPS 2024**: Annual Conference on Neural Information Processing Systems

[C6] **Chengpeng Wang**, Wuqi Zhang, Zian Su, Xiangzhe Xu, and Xiangyu Zhang, Sanitizing Large Language Models in Bug Detection with Data-Flow, In **Findings of EMNLP 2024**: Empirical Methods in Natural Language Processing

[C7] Li Lin, Zongyin Hao, **Chengpeng Wang**, Zhuangda Wang, Rongxin Wu, and Gang Fan, SQLess: Dialect-Agnostic SQL Query Simplification, In **ISSTA 2024**: International Symposium on Software Testing and Analysis

[C8] **Chengpeng Wang**, Jipeng Zhang, Rongxin Wu, and Charles Zhang, DAInfer: Inferring API Aliasing Specifications from Library Documentation via Neurosymbolic Optimization, In **FSE 2024**: International Conference on the Foundations of Software Engineering

[C9] Bowen Zhang, Wei Chen, Peisen Yao, **Chengpeng Wang**, Wensheng Tang, and Charles Zhang, SIRO: Empowering Version Compatibility in Intermediate Representations via Program Synthesis, In **ASPLOS 2024**: ACM Conference on Architectural Support for Programming Languages and Operating Systems

[C10] Hao Ling, Heqing Huang, **Chengpeng Wang**, Yuandao Cai, and Charles Zhang, GiantSan: Efficient Memory Sanitization with Segment Folding, In **ASPLOS 2024**: ACM Conference on Architectural Support for Programming Languages and Operating Systems **(ACM SIGARCH Best Paper Award)**

[C11] Rongxin Wu, Yuxuan He, Jiafeng Huang, **Chengpeng Wang***, Wensheng Tang, Qingkai Shi, Xiao Xiao, and Charles Zhang, LibAlchemy: A Two-Layer Persistent Summary Design for Taming Third-Party Libraries in Static Bug-Finding Systems, In **ICSE 2024**: International Conference on Software Engineering

[C12] Wensheng Tang[#], **Chengpeng Wang**[#], Peisen Yao, Rongxin Wu, Xianjin Fu, Gang Fan, and Charles Zhang, DCLink: Bridging Data Constraint Changes and Implementations in FinTech Systems, In **ASE 2023** : International Conference on Automated Software Engineering

[C13] **Chengpeng Wang**, Peisen Yao, Wensheng Tang, Gang Fan, Charles Zhang, Synthesizing Conjunctive Queries for Code Search, In **ECOOP 2023**: European Conference on Object-Oriented Programming

[C14] Zongyin Hao, Quanfeng Huang, **Chengpeng Wang**, Jianfeng Wang, Yushan Zhang, Rongxin Wu, and Charles Zhang, Pinolo: Detecting Logical Bugs in Database Management Systems with Approximate Query Synthesis, In **ATC 2023**: USENIX Annual Technical Conference

[C15] **Chengpeng Wang**, Gang Fan, Peisen Yao, Fuxiong Pan, and Charles Zhang, Verifying Data Constraint Equivalence in FinTech Systems, In **ICSE 2023**: International Conference on Software Engineering

[C16] Rongxin Wu, Minglei Chen, **Chengpeng Wang***, Gang Fan, Jiguang Qiu, and Charles Zhang, Accelerating Build Dependency Error Detection via Virtual Build, In **ASE 2022** : International Conference on Automated Software Engineering

[C17] **Chengpeng Wang**, Peisen Yao, Wensheng Tang, Qingkai Shi, and Charles Zhang: Complexity-Guided Container Replacement Synthesis, In **OOPSLA 2022**: SIGPLAN Conference on Objected Oriented Programming, Systems, Languages and Applications **(ACM SIGPLAN Distinguished Paper Award)**

[C18] Gang Fan, **Chengpeng Wang**, Rongxin Wu, Xiao Xiao, Qingkai Shi, and Charles Zhang: Escaping Dependency Hell: Finding Build Dependency Errors with the Unified Dependency Graph, In **ISSTA 2020**: International Symposium on Software Testing and Analysis

**WORKSHOP and POSTER PUBLICATIONS**

[W1] Ming Liang, Xiaoheng Xie, Gehao Zhang, Xunjin Zheng, Peng Di, Wei Jiang, Hongwei Chen, **Chengpeng Wang**, and Gang Fan, RepoGenix: Dual Context-Aided Repository-Level Code Completion with Language Models, In Poster Track, **ASE 2024**: International Conference on Automated Software Engineering

[W2] **Chengpeng Wang**, CodeSpider: Automatic Code Querying with Multi-modal Conjunctive Query Synthesis, In SRC track, **SPLASH 2022**: SIGPLAN conference on Systems, Programming, Languages, and Applications: Software for Humanity, Student Research Competition

[W3] **Chengpeng Wang**, Yixiao Yang, Han Liu, and Le Kang: Statistical API Completion Based on Code Relevance Mining, In **MAINT 2019**: International Workshop on Mining and Analyzing Interaction Histories

## ACADEMIC SERVICES

**Program Organizing Committee Member:**

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|---|---|
| International Workshop on Language Models and Programming Languages (LMPL) | *2025* |

**Program Committee Member:**

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| Annual Conference on Neural Information Processing Systems (NeurIPS) | *2025* |
| International Conference on Automated Software Engineering (ASE) | *2025* |
| International Conference on AI Foundation Models and Software Engineering (Forge) | *2024, 2025* |
| International Conference on the Foundations of Software Engineering (FSE) | *2024,* 2025 |
| International Symposium on Software Testing and Analysis (ISSTA) | *2025* |
| SPLASH, Student Research Contest Track | *2024* |
| International Conference on Automated Software Engineering (ASE), Industrial Track | *2024* |
| International Symposium on Software Reliability Engineering (ISSRE) | *2024, 2025* |

**Journal Reviewer:**

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| Transactions on Software Engineering (TSE) | 2024 |

**Artifact Evaluation Committee Member:**

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| International Conference on OOP, Systems, Languages, and Applications (OOPSLA) | *2024* |
| ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI) | *2023* |
| International Conference on the Foundations of Software Engineering (FSE) | *2022* |
| International Symposium on Software Testing and Analysis (ISSTA) | *2022* |

**Volunteer:**

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| International Conference on OOP, Systems, Languages, and Applications (OOPSLA) | *2022* |
| Student Volunteer at International Symposium on Software Testing and Analysis (ISSTA) | *2019* |
| Long-term mentor of SIGPLAN-M | *2024, 2025* |

## TEACHING

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| CS 592: AI and Security, Purdue University | *Fall 2024* |
| COMP 3021: Java Programming, HKUST | *Spring/Fall 2022/2023* |
| COMP 4631: Computer and Communication Security, HKUST | *Fall 2021* |
| COMP 3111/H: Software Engineering, HKUST | *Fall 2020* |
| COMP 2011: Programming with C++, HKUST | *Spring 2020* |
| Haskell: Functional Language Programming, THU | *Spring 2019* |
| Automaton and Formal Logic, THU | *Fall 2018* |
| Drop-in Tutoring for STEM Courses, THU | *Spring/Fall 2018/2019* |

## ADVISING AND MENTORING

**Jinyao Guo (Graduate Student, Purdue University)** *Nov 2024 - Present*
*Currently a Ph.D. student at Purdue University*
Working on LLM-driven code auditing and coauthored a publication at ICML 2025 [C1]

**Mingwei Zheng (Graduate Student, Purdue University)** *Aug 2024 - Present*
*Currently a Ph.D. student at Purdue University*

Working on network protocol bug detection and coauthored a publication at ISSTA 2025 [C3] and the preprint [P1].

**Yifei Gao (Graduate Student, Purdue University)** *May 2024 - Present*
*Currently a Ph.D. student at Purdue University*
Working on automated translation from C to safer Rust and coauthored the preprint [P3].

**Pengxiang Huang (Graduate Student, Northwestern University)** *May 2024 - Present*
*Currently a Master student at Northwestern University*
Working on automated translation from C to safer Rust and coauthored the preprint [P3].

**Bowen Zhang (Graduate Student, HKUST)** *Oct 2022 - Dec 2023*
*Currently a Ph.D. student at Hong Kong University of Science and Technology*
Worked on IR Translator Synthesis and coauthored a publication at ASPLOS 2024 [C9].

## PRESENTATIONS AND INVITED TALKS

**RepoAudit: Human-like AI Auditor for Code Repositories**
RSA Conference 2025, San Francisco — Apr 2025

**Advances in AI-powered Code Security: Next-Level Bug Detection**
Uber's Programming Systems Team. Virtual — Feb 2025
CodeQL@GitHub. Virtual — Jan 2025

**Neuro-Symbolic Static Analysis for Reliable Software Systems**
UC Davis, UIUC, UNSW — May 2025
AST Lab, ETH Zurich. Virtual — Mar 2025
ByteDance. Virtual — Jan 2025
Department of Computing, Hong Kong Polytechnic University — Jan 2025
Institute of Data Science, The University of Hong Kong — Dec 2024

**Neuro-Symbolic Static Analysis for Reliable and Performant Software Systems**
School of Computer Science, Nanjing University. Nanjing, China — Dec 2024

**LLMDFA: Analyzing Dataflow in Code with Large Language Models**
Ant Group. Virtual — Dec 2024

**When Static Analysis Meets Large Language Models: A Neuro-Symbolic Approach**
Boston Computation Club. Virtual — Nov 2024
AI-CyberSecurity Research Lunch Talk, Texas A&M University (TAMU). Virtual — Oct 2024
School of Software, Tsinghua University. Virtual — Oct 2024

**Towards Enhancing Reliability and Performance of Data-Centric Systems with Static Analysis**
School of Informatics, Xiamen University. Xiamen, China — Aug 2023

**Synthesizing Conjunctive Queries for Code Search**
ByteDance. Virtual — Jun 2023

**Complexity-Guided Container Replacement Synthesis**
AST Lab, ETH Zurich. Virtual — Mar 2023